SMART POLYPHARMACY AND VITAMIN RECOMMENDATION SYSTEM FOR GERIATRIC PATIENTS

25-26J-368

Project Final Report

Tharaka B.W.P

IT22600516

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

May 2026

PERSONALIZED MENTAL HEALTH AND LIFESTYLE ADVISORY SYSTEM FOR GERIATRIC PATIENTS

25-26J-368

Project Final Report

Tharaka B.W.P
IT22600516

Supervisor – Ms. Jenny Krishara
Co-Supervisor - Dr. Dinuka R Wijendra

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

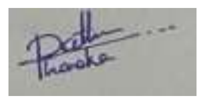
Sri Lanka Institute of Information Technology
Sri Lanka

May 2026

DECLARATION

We declare that this is our own work, and this report does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

DECLARATION OF STUDENT

Name	Student ID	Signature	Date
Tharaka B.W.P	IT22600516		04/26/2026

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor
Ms. Jenny Krishara

Date : 04/26/2026

ACKNOWLEDGMENT

The completion of this research project marks an important milestone in my academic journey, made possible through the support, guidance, and encouragement of many individuals. I would like to express my sincere gratitude to everyone who contributed to the success of this work.

First and foremost, I would like to extend my heartfelt appreciation to my supervisor, Ms. Jenny Krishara, and co-supervisor, Dr. Dinuka R. Wijendra, for their continuous guidance, valuable insights, and constructive feedback throughout this research. Their expertise and encouragement greatly assisted me in refining my ideas and successfully completing this study.

I would also like to express my gratitude to the Dean, Prof. Pradeep Abeygunawardhana, the Head of the Department of Information Technology, Prof. Samantha Rajapaksha, and the module lecturer of the Research Project, Prof. Koliya Pulasinghe, along with the academic staff of the Sri Lanka Institute of Information Technology (SLIIT), for providing a supportive and resourceful academic environment. Their dedication to academic excellence has been a strong foundation for this research.

I would like to thank my research team members for their collaboration, teamwork, and shared commitment throughout this project. Working together has helped overcome challenges and achieve our research goals effectively.

Finally, I would like to express my deepest gratitude to my family for their unconditional love, patience, and encouragement. Their continuous support and belief in me have been my greatest strength throughout my academic journey.

Abstract

The AI Lifestyle & Mental Wellness Advisor is an intelligent system designed to support geriatric patients, especially those managing multiple medications, by providing non-pharmacological and non-medical daily guidance for stress reduction, emotional stability, and healthy living. Unlike existing wellness apps that focus solely on reminders or general advice, this system integrates polypharmacy risk scores, patient-reported mental health data, and facial expression analysis to generate personalized, risk-aware recommendations.

The platform delivers practical, drug-free interventions such as mindfulness exercises, relaxation techniques, proper sleep hygiene, hydration tracking, and safe physical activity, tailored to the patient's overall health and medication burden. It also performs continuous monitoring to identify warning signs of severe distress—such as suicidal thoughts, depression, or anxiety spikes—and immediately directs the patient to seek professional medical assistance. Through motivational check-ins, emotional support, and lifestyle coaching, the advisor promotes independence, safety, and well-being among elderly users.

This research is innovative because it is the first system to fuse polypharmacy risk scoring, self- reported mental health status, and facial expression monitoring into a unified AI companion for geriatric care. The approach employs AI-based personalization, lightweight interfaces for accessibility, and multi-modal mental health indicators for accurate risk assessment. Expected outcomes include improved mood, reduced stress, healthier daily routines, and enhanced quality of life for elderly patients managing complex medication regimens.

Keywords: Geriatric care, Polypharmacy, AI companion, Mental wellness, Lifestyle guidance, Drug-free intervention

Table of Contents

LIST OF FIGURES

LIST OF TABLES

LIST OF APPENDIX

LIST OF ABBREVIATIONS

Abbreviation	**Description**
AI	Artificial Intelligence
SLIIT	Sri Lanka Institute of Information Technology
ICT	Information and Communication Technology
CBT	Cognitive Behavioral Therapy
NLP	Natural Language Processing
PSS	Perceived Stress Scale
GDS	Geriatric Depression Scale
ML	Machine Learning
CNN	Convolutional Neural Network
API	Application Programming Interface
SUS	System Usability Scale
MTCNN	Multi-task Cascaded Convolutional Network
B2B2C	Business-to-Business-to-Consumer
NLP	Natural Language Processing

Table 1 : List of abbreviations

1 INTRODUCTION

The global demographic landscape is undergoing a profound transformation. The population aged 60 and above is expected to nearly double by 2050, reaching approximately 2 billion individuals. While increased life expectancy reflects major advancements in healthcare and living standards, it also presents new challenges. Aging is often accompanied by a rise in chronic, noncommunicable diseases such as diabetes, cardiovascular conditions, arthritis, and dementia. Managing these multiple conditions frequently requires **polypharmacy**, defined as the concurrent use of five or more medications.

Elderly patients become more vulnerable to adverse drug reactions, harmful drug–drug interactions, reduced adherence to treatment, and increased rates of hospitalization. It is observed that both these physical health risks and the associated psychological and emotional challenges often receive comparatively less systematic attention. The cumulative burden of managing multiple medications, along with chronic illness, can contribute to stress, anxiety, and reduced overall well-being, highlighting the need for a more integrated approach to patient care.

A considerable proportion of older adults silently struggle with psychological challenges. Studies indicate that around 15–20% of elderly individuals experience anxiety or depression, while many others suffer from chronic sleep disturbances. These mental health issues are not isolated concerns they directly influence physical health by accelerating disease progression, impairing cognitive function, and reducing independence and quality of life. Unfortunately, due to social stigma, limited awareness, and fragmented healthcare systems, these emotional struggles often go undetected and untreated.

In response to these challenges, digital health technologies have emerged to support elderly care. However, most existing solutions are heavily medication-focused, offering features such as dosage reminders and adherence tracking. While useful, these systems fail to address a critical aspect of geriatric care the strong connection between medication burden, emotional well-being, and daily lifestyle habits. As a result, there remains a significant gap in delivering holistic and patient-centered care for elderly individuals.

This gap highlights the urgent need for a more comprehensive and integrated approach. Rather than focusing solely on medications, there is a growing demand for intelligent systems that consider the full spectrum of an elderly patient’s well-being physical, emotional, and behavioral.

To address this need, this research proposes the Lifestyle & Mental Wellness Advisor, an intelligent and personalized system designed specifically for geriatric patients managing complex medication regimens. Unlike traditional applications, this system goes beyond reminders by combining polypharmacy risk assessment, self-reported mental health data, and facial expression analysis to generate tailored, risk-aware lifestyle recommendations.

The proposed system emphasizes non-pharmacological interventions such as mindfulness practices, relaxation techniques, sleep hygiene, hydration monitoring, and safe physical activity. Additionally, it continuously monitors users to detect early signs of psychological distress, including anxiety, depression, or severe emotional changes, and provides timely guidance to seek professional help when necessary.

By integrating multiple dimensions of patient data into a single platform, this research introduces a novel, holistic approach to elderly care. The system aims not only to improve medication safety but also to enhance emotional well-being, promote independence, and ultimately improve the overall quality of life for older adults.

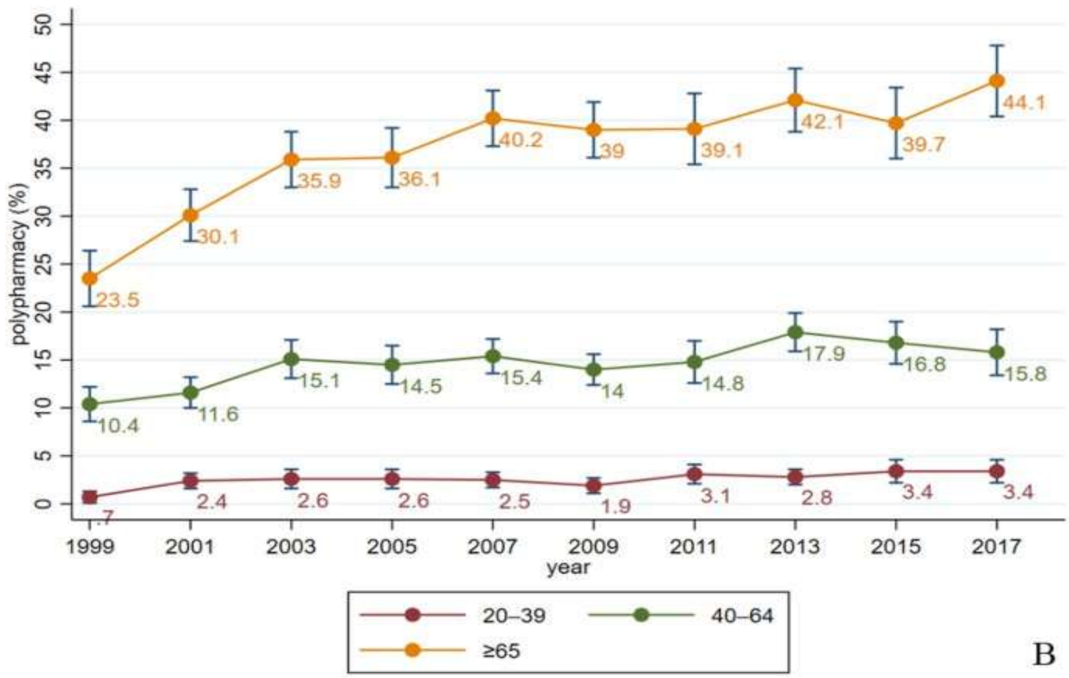


Figure 1 : Prevalence and trend of polypharmacy by age [13]

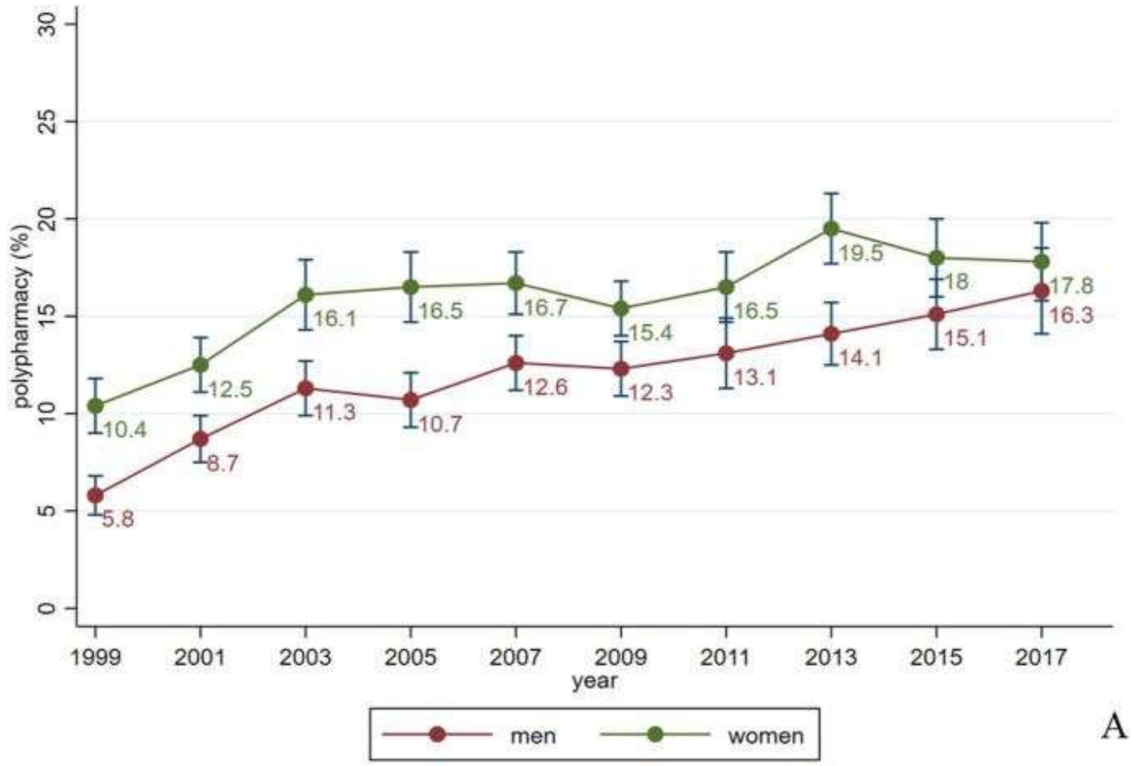


Figure 2 : Prevalence and trend of polypharmacy by sex [13]

1.1. Background & literature survey

With the rapid growth of the elderly population, managing health in later life has become increasingly complex and demanding. Many older adults are required to take multiple medications while simultaneously coping with chronic illnesses, lifestyle changes, and emotional challenges. In recent years, research has started to move beyond purely clinical treatment approaches and has begun to explore more patient-centered, technology-driven solutions. Several studies have focused on areas such as medication risk assessment, nutritional support, and detection of drug-related complications. Building on these developments, integrated digital health systems are now being explored to address multiple aspects of elderly care within a single platform.

In line with this direction, the present study is part of a broader system developed collaboratively to support geriatric patients through different but interconnected components. While other components focus on areas such as identifying medication-induced vitamin deficiencies, generating personalized meal plans, and assessing polypharmacy risk, this research specifically concentrates on lifestyle and mental wellness support. By focusing on non-pharmacological guidance and emotional well-being, this component aims to complement existing clinical and technical solutions, contributing to a more holistic and patient-centered approach to elderly care.

1.1.1 The Dual Burden of Polypharmacy and Mental Health in Aging Populations

The global shift toward an aging population is accelerating at an unprecedented rate. By 2050, the number of individuals aged 60 years and above is expected to reach nearly 2 billion, effectively doubling the current population [1]. As people age, the likelihood of living with multiple chronic, noncommunicable diseases such as hypertension, diabetes, arthritis, and neurodegenerative disorders also increases. Managing these coexisting conditions often requires the use of multiple medications, commonly referred to as polypharmacy, typically defined as the concurrent use of five or more drugs [2]. While this approach is often clinically necessary, it is associated with several risks, including adverse drug reactions (ADRs), harmful drug–drug interactions, reduced medication adherence, and increased rates of preventable hospitalizations [3].

Alongside these physical health challenges, older adults also experience a significant burden of mental health concerns. Epidemiological studies suggest that approximately 15 - 20% of elderly individuals suffer from clinically relevant anxiety or depression, while a substantial proportion report persistent sleep disturbances [4]. These conditions are closely linked with physical health, often worsening disease outcomes, increasing dependency, and contributing to cognitive decline. However, factors such as social stigma, limited access to specialized geriatric mental health services, and a stronger clinical focus on physical symptoms can lead to under-recognition and under-treatment of these issues.

Importantly, polypharmacy and mental health are closely interconnected. Certain medications or combinations of drugs may contribute to or exacerbate symptoms such as anxiety, depression, and sleep disorders. At the same time, untreated psychological distress can negatively impact medication adherence, creating a cycle that further complicates patient outcomes. Previous studies, including Wang et al. (2023), have shown that the prevalence of polypharmacy continues to rise, particularly among individuals aged over 70 years [13]. This dual burden highlights the need for integrated approaches that address both medication-related risks and mental well-being an area that remains insufficiently addressed in current healthcare systems.

1.1.2 Evolution of Digital Health Tools for Geriatric Care

The initial generation of digital health solutions for older adults primarily focused on medication management and adherence. Platforms such as MedMinder and PillPack were designed to organize medication schedules, automate refills, and provide reminders for missed doses. While these tools significantly improved adherence, their scope remained limited, as they largely addressed medication-taking behavior without considering the broader emotional and lifestyle needs of elderly patients.

Subsequently, a second wave of digital health interventions emerged to address non-clinical aspects of aging. These included mobile applications and community-based programs targeting sleep quality, stress management, and physical activity. Interventions such as cognitive behavioral therapy (CBT) for insomnia, mindfulness-based stress reduction, and structured exercise programs have demonstrated positive outcomes among older populations. For instance, Rusch et al. (2019) reported that mindfulness-based interventions resulted in moderate to significant improvements in sleep quality, with sustained benefits over time [6]. Similarly, non-pharmacological approaches involving psychological support, social engagement, and light physical activity have been shown to reduce loneliness and depressive symptoms in community-dwelling elderly individuals [12].

Despite these advancements, existing solutions often operate in isolation. Many tools focus on a single aspect of care such as medication adherence, physical activity, or mental wellness without integrating these dimensions into a unified system. As a result, they fail to adapt recommendations based on an individual's medication burden, emotional state, or overall health risk profile. This fragmentation limits their effectiveness and contributes to gaps in delivering truly personalized and holistic care.

Therefore, there is a growing need for integrated, intelligent systems that can combine multiple sources of patient information to provide context-aware support. In response to this gap, the present study incorporates multi-modal inputs, including facial expression analysis and patient-reported behavioral patterns, to better understand emotional states and deliver personalized, non-pharmacological lifestyle recommendations. This approach aims to bridge the divide between clinical management and everyday well-being, offering a more comprehensive solution for geriatric care.

1.1.3 Limitations of Existing Geriatric Care Systems

Although significant progress has been made in digital health technologies for older adults, many existing systems still operate in a fragmented and task-specific manner. Most solutions tend to focus on isolated aspects of care, such as medication reminders, appointment scheduling, or general wellness tracking. While these features are helpful in improving routine management, they often fail to capture the broader and more interconnected nature of geriatric health.

In real-world scenarios, elderly patients rarely experience health challenges in isolation. Medication regimens, emotional well-being, dietary habits, and lifestyle behaviors are closely interlinked. However, current systems generally do not integrate these dimensions into a unified view of the patient. As a result, healthcare support often becomes reactive rather than proactive, addressing individual issues separately instead of considering the overall health context.

Another key limitation is the lack of personalization based on combined risk factors. For instance, medication-related risks and nutritional implications are often assessed independently, without being connected to the patient's emotional state or daily behavior patterns. This separation can lead to incomplete guidance, where important relationships between physical health, mental well-being, and lifestyle are overlooked. These gaps highlight the need for more integrated systems that can bring together multiple aspects of patient information in a meaningful and usable way.

1.1.4 Need for Integrated Patient-Reported and Risk-Aware Support Systems

In response to the limitations of existing approaches, there is a growing need for healthcare systems that emphasize integration, continuity, and patient participation. Older adults, especially those managing multiple medications, require support that reflects not only their clinical conditions but also their daily experiences, emotional state, and lifestyle habits.

One of the key emerging approaches in this context is the use of patient-reported data as a central component of care. Information such as mood variations, sleep quality, energy levels, and daily activity patterns can provide valuable insight into a patient's overall well-being. When combined with clinical indicators such as polypharmacy risk scores and medication-related nutritional concerns, this data can help form a more complete and realistic understanding of the patient's health status.

This integrated perspective allows for more meaningful and timely support. Instead of relying solely on isolated clinical assessments, healthcare systems can better align recommendations with the patient's current condition and needs. For example, identifying potential medication-induced deficiencies alongside lifestyle patterns can help guide appropriate laboratory investigations and preventive care strategies.

Overall, this highlights the importance of developing systems that connect different dimensions of elderly care into a single coherent framework. Such approaches aim to move beyond fragmented monitoring toward more holistic, context-aware support that is both practical and responsive to individual patient needs.

1.2 Literature review

1.2.1 Polypharmacy and Its Indirect Impact on Patient Well-being

Polypharmacy is commonly defined as the use of five or more medications at the same time, and it is widely observed among older adults who suffer from multiple chronic conditions. While these medications are often necessary for disease control, managing several drugs daily can become complex and physically demanding for patients. This complexity can lead to confusion, missed doses, or incorrect usage, which in turn reduces treatment effectiveness.

Beyond the clinical risks such as adverse drug reactions and drug–drug interactions, polypharmacy also affects the overall well-being of patients in more indirect ways. Elderly individuals often experience increased stress and fatigue due to managing complicated medication schedules. Over time, this can reduce their confidence in self-care and affect their daily lifestyle routines. As a result, polypharmacy not only influences physical health but also contributes to a decline in general quality of life.

1.2.2 Medication-Induced Nutritional Deficiencies and Clinical Implications

Long-term medication use has been shown to interfere with the absorption and metabolism of essential nutrients in the body. Certain commonly prescribed drugs in elderly care can lead to deficiencies in vitamins such as Vitamin B12, Vitamin D, and minerals like calcium and iron. These deficiencies may not appear immediately but can develop gradually over time.

The clinical impact of such deficiencies can be significant. They may contribute to symptoms such as fatigue, muscle weakness, bone fragility, cognitive decline, and reduced immunity. In many cases, these symptoms overlap with normal aging, making early identification more difficult. Laboratory testing is therefore an important tool for confirming suspected deficiencies and guiding appropriate treatment.

However, in many healthcare settings, testing is often performed only after symptoms become visible. This reactive approach may delay early intervention. This highlights the importance of systems that can help identify potential risk factors earlier and guide timely laboratory investigations.

1.2.3 Lifestyle Interventions in Elderly Care

Lifestyle interventions play a crucial role in maintaining health and well-being among older adults. Simple daily practices such as maintaining proper sleep hygiene, staying hydrated, engaging in light physical activity, and practicing relaxation techniques can significantly improve both physical and mental health outcomes.

Research has shown that non-pharmacological approaches can help reduce stress, improve sleep quality, and support better management of chronic conditions. Unlike medication-based treatments, lifestyle interventions are generally low-risk and can be adapted according to an individual's physical capability and living conditions.

However, most general lifestyle recommendations available today are not personalized. Elderly individuals often receive the same advice regardless of their medical conditions, medication burden, or emotional state. This limits the effectiveness of such interventions and highlights the need for more tailored lifestyle guidance based on individual health profiles.

1.2.4 Patient-Reported Data in Behavioral and Emotional Health Monitoring

Patient-reported data refers to information directly provided by individuals about their own health status, daily habits, and emotional experiences. In elderly care, this may include self-reported mood, sleep patterns, energy levels, and lifestyle behaviors.

This type of data is valuable because it provides insights that may not always be captured through clinical tests alone. For example, changes in mood or sleep quality can indicate early signs of stress, anxiety, or declining health. When collected consistently, patient-reported information can help healthcare systems better understand the day-to-day condition of individuals.

However, this type of data also has limitations. It is subjective and may vary depending on the patient's perception, memory, or willingness to report accurately. Therefore, its effectiveness improves significantly when it is combined with clinical or risk-based information rather than used in isolation.

1.2.5 Clinical Decision Support for Laboratory Testing in Preventive Care

Laboratory tests play an essential role in identifying and confirming nutritional deficiencies, especially in older adults. Tests for vitamin levels, blood counts, and metabolic markers help clinicians detect underlying conditions that may not yet show clear physical symptoms.

In traditional healthcare practice, laboratory testing is usually performed based on visible symptoms or routine check-ups. While this approach is effective, it may not always support early detection of gradual or hidden deficiencies. As a result, some conditions are identified only after they have progressed significantly.

There is increasing interest in improving this process through more informed and preventive approaches to testing. By linking patient health profiles, medication history, and reported symptoms, it becomes possible to better identify individuals who may benefit from specific laboratory investigations. This helps shift care from a reactive model to a more preventive and timely approach.

1.3 Research Gap

A review of existing literature and available digital health solutions indicates that current approaches to elderly care are largely fragmented and lack integration across key domains. While significant work has been done in areas such as polypharmacy risk assessment, nutritional deficiency detection, and lifestyle intervention, these components are often addressed independently rather than as part of a connected system.

In particular, there is limited focus on combining patient-reported emotional and lifestyle data with medication-related risk information to guide practical, everyday support. Most existing systems do not utilize simple, continuously collected inputs—such as mood, sleep patterns, and daily habits—to adjust recommendations in a meaningful way. As a result, the support provided is often generalized and not tailored to the individual's current condition.

Additionally, although medication-induced nutritional deficiencies are well documented, there is a lack of systems that link these risks with timely and context-specific laboratory test recommendations. Current practices tend to rely on reactive testing based on visible symptoms, rather than proactive guidance informed by combined health indicators.

Therefore, there exists a clear gap in developing a patient-centered, integrated support approach that connects lifestyle guidance, emotional well-being, and clinical follow-up considerations. Addressing this gap can contribute to more practical, personalized, and preventive care for elderly individuals managing complex medication regimens.

1.4 Research Problem

The rapid growth of the elderly population has led to an increasing number of individuals managing multiple chronic conditions simultaneously. As a result, polypharmacy has become a common practice in geriatric care. While necessary for disease management, the use of multiple medications often places a significant burden on older adults, affecting not only their physical health but also their daily routines, emotional well-being, and overall quality of life.

At the same time, many elderly patients experience challenges such as poor sleep, stress, reduced physical activity, and changes in mood. These factors are closely linked with their health status but are not always systematically monitored in routine care. In addition, long-term medication use may contribute to nutritional deficiencies, which often remain undetected until symptoms become more pronounced and require clinical investigation.

Although various digital tools and healthcare systems exist, most of them focus on isolated aspects of care, such as medication reminders, dietary advice, or general wellness tracking. These approaches often do not consider the combined impact of medication burden, lifestyle behaviors, and emotional state. As a result, elderly patients may receive fragmented support that does not fully address their day-to-day needs.

Furthermore, decisions regarding laboratory testing for potential deficiencies are typically based on clinical symptoms rather than integrated patient data. This can delay early identification and intervention. Therefore, there is a need for a more coordinated approach that brings together patient-reported information, medication-related risks, and preventive care strategies.

1.5 Objectives

1.5.1 Primary Objective

The primary objective of this research is to design and develop a lifestyle and mental wellness support system that provides non-pharmacological guidance for elderly individuals managing multiple medications. The system focuses on improving day-to-day well-being by combining patient-reported emotional and lifestyle data with medication-related risk information obtained from a polypharmacy risk module.

In addition, the system incorporates simple emotion detection through facial expression analysis together with self-reported inputs to better understand the user's current state. Based on this combined information, the system delivers personalized lifestyle recommendations, stress management support, and guidance for appropriate laboratory investigations related to medication-induced nutritional deficiencies.

The system is intended to function as a supportive and preventive tool that complements existing healthcare services, helping elderly users maintain healthier routines, improve emotional well-being, and seek timely medical advice when necessary.

1.5.2 Specific Objectives

1. To incorporate medication-related risk information into lifestyle guidance

This objective focuses on utilizing the polypharmacy risk score generated by an external module as an input to the system. The lifestyle recommendations provided—such as activity levels, relaxation practices, and daily routines—are adjusted according to the patient's level of medication-related risk. This ensures that the guidance remains safe, appropriate, and relevant to the individual's overall condition.

2. To collect and interpret patient-reported emotional and lifestyle data

A structured approach will be used to gather information such as mood, stress levels, sleep quality, physical activity, and hydration habits. In addition to these self-reported inputs, basic facial expression analysis is incorporated to provide supportive insight into the user's emotional state. These combined inputs help to form a more complete understanding of the individual's daily well-being.

3. To provide personalized, non-pharmacological lifestyle recommendations

Based on the collected data and associated risk information, the system generates tailored lifestyle suggestions. These include stress reduction techniques, sleep hygiene practices, light physical activity recommendations, and hydration guidance. The aim is to support healthier daily habits without introducing additional medications.

4. To support early identification of potential nutritional deficiencies through lab test guidance

Using information related to medication-induced vitamin deficiencies (identified by a separate module), the system provides suggestions for relevant laboratory tests. This helps encourage timely clinical investigation and supports a more preventive approach to managing potential deficiencies.

5. To implement continuous lifestyle monitoring and user engagement

The system includes regular check-ins and simple daily interactions to encourage user participation. These may include reminders, motivational prompts, and routine tracking, which help maintain consistency in healthy behaviors and promote a sense of engagement in daily self-care.

6. To identify signs of emotional distress and encourage appropriate action

The system continuously reviews patient-reported inputs and observed emotional indicators to detect patterns that may suggest increased stress or emotional difficulty. When such patterns are identified, the system provides gentle guidance encouraging the user to seek support from a healthcare professional or a trusted caregiver, without attempting to replace professional care.

1.5.3 Scope and Delimitations

This single component specifically focuses on providing lifestyle and mental wellness support for individuals managing multiple medications. It is designed to collect and interpret patient-reported information related to emotional state and daily lifestyle patterns, including mood, stress levels, sleep quality, physical activity, and hydration habits. In addition, basic facial expression analysis is incorporated to provide supportive insight into the user's emotional condition. These inputs are combined with medication-related risk information obtained from an external polypharmacy risk module, enabling the system to deliver personalized, non-pharmacological lifestyle recommendations. Furthermore, this component provides guidance on appropriate laboratory tests based on medication-induced nutritional deficiencies identified by another module within the overall system.

This single component is intended to function as a supportive and preventive tool rather than a clinical decision-making system. The recommendations generated are limited to lifestyle improvements, stress management, and general well-being, and are not intended to diagnose or treat medical conditions. Similarly, the suggested laboratory tests are provided only as guidance to encourage timely consultation with healthcare professionals. The system relies on user-reported data, which may be subjective, and the facial expression analysis is used only as a supplementary indicator rather than a definitive measure of emotional state.

This study is specifically focused on how this single component integrates the outputs with patient-reported information to provide meaningful and practical lifestyle guidance. Limitations such as variability in user input, differences in individual health conditions, and the basic nature of emotion detection may influence the effectiveness of the system.

2 METHODOLOGY

2.1 PROJECT REQUIREMENTS

The development of the lifestyle and mental wellness counseling system is guided by a comprehensive set of functional, non-functional, user, and system requirements. These requirements ensure that the system integrates multiple medication risk, emotional state, and nutritional deficiency data into a safe, personalized, non-medical guide for older patients.

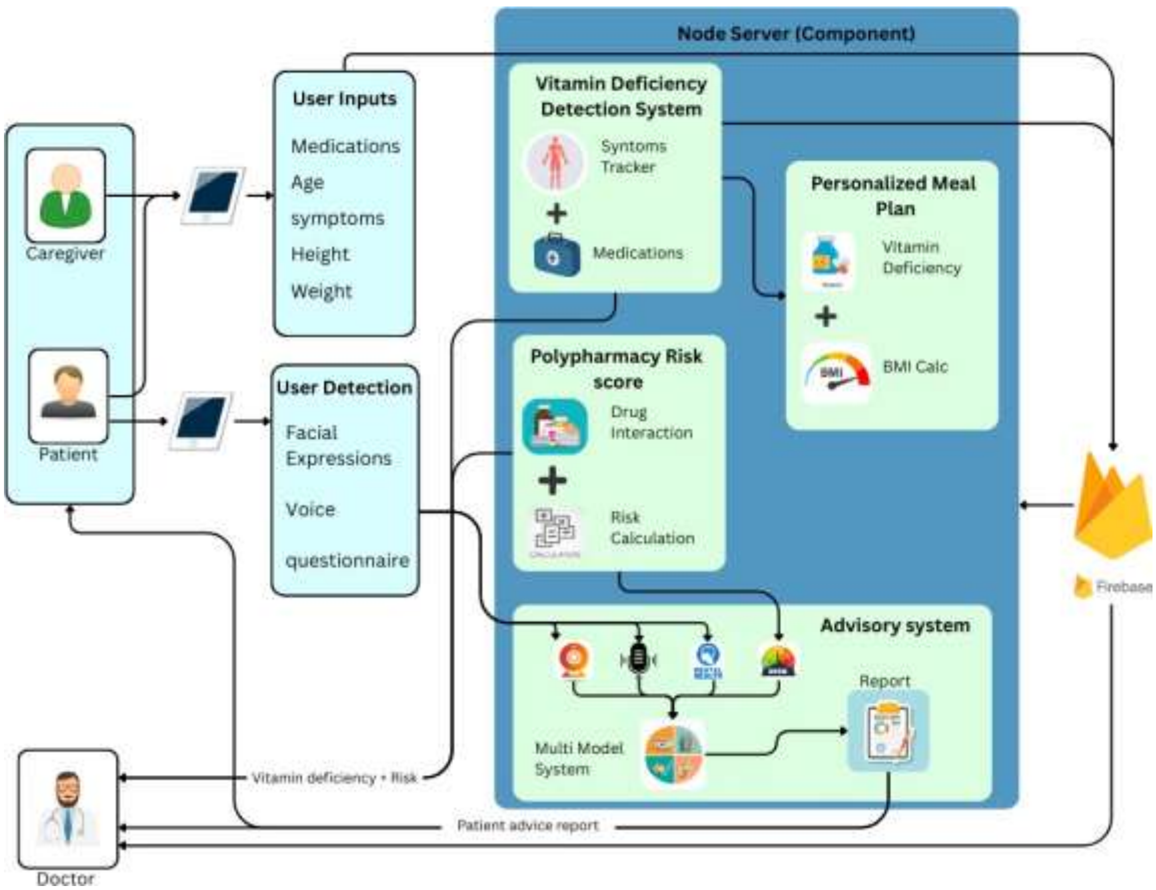


Figure 3 : Overall System Overview Diagram

2.2 Functional Requirements

- Patient-Reported Data Collection: Patients can log daily routines, stress levels, sleep quality, mood, and activity patterns via structured checklists, sliders, or voice input. Voice-based interaction is prioritized to assist seniors with limited digital literacy.

- Facial Expression Recognition: The AI system must analyze facial expressions in real-time to detect mood states such as stress, sadness, or low engagement, complementing self-reported data.
- Personalized Lifestyle Guidance: Using machine learning, the system must generate lifestyle advice tailored to each user's health profile, polypharmacy risk score, and emotional state (e.g., sleep hygiene, hydration reminders, gentle exercises).
- Mental Wellness Recommendations: The platform should provide non-pharmacological interventions like mindfulness exercises, journaling prompts, relaxation practices, and breathing routines, dynamically adjusted based on mood/stress input.
- Daily Motivational Check-ins: The companion must deliver empathetic reminders, positive affirmations, and motivational interactions to sustain patient engagement and reduce loneliness.
- Risk Detection and Safety Alerts: AI models must flag warning signs of severe mental distress (e.g., suicidal ideation, acute depression) using psychological scoring and emotional signals. With prior consent, alerts may also notify caregivers or family members.

2.3 Non-Functional Requirements

Non-functional requirements describe the qualities and constraints of the system, focusing on performance, reliability, and safety.

- **Accuracy:** The AI system must provide highly accurate recommendations and risk assessments based on validated models and patient-reported data [8].
- **Real-Time Responsiveness**: Recommendations, motivational check-ins, and risk alerts should be delivered in real-time, allowing immediate action when necessary [9].
- **Usability:** The interface must be intuitive and simple, allowing elderly patients to navigate the system without extensive training. Graphical indicators, easy-to-read fonts, and minimalistic design should be used.
- **Security and Privacy:** All patient data must be encrypted during transmission and storage. Role-based access control must be implemented to prevent unauthorized access [10].

- **Scalability:** The system must support multiple concurrent users, handling long-term data storage and analysis without degradation in performance.
- **Compliance:** The system must comply with data protection regulations such as GDPR, HIPAA, or local healthcare privacy standards, ensuring ethical handling of sensitive patient data.
- **Reliability and Availability:** The system should be operational 24/7, with minimal downtime, ensuring that patients can access support whenever needed.

2.4 User Requirements

User requirements ensure the system is **accessible, safe, and supportive**:

Elderly Patients:

- A simple, minimalistic interface with large fonts, clear icons, and voice-based input/output.
- Accessible across different literacy levels and adaptable for cognitive decline.
- Privacy-preserving, ensuring sensitive lifestyle and emotional data remain confidential.

Caregivers and Family:

- With patient consent, authorized caregivers can access summary reports and distress alerts.
- Ability to receive real-time notifications in emergencies.

Cross-Platform Accessibility:

- Application must work seamlessly on smartphones, tablets, and desktops.
- Offline features (local caching) for patients with inconsistent internet access.

2.5 System and Software Requirements

Hardware Requirements (User Side): A smartphone, tablet, or laptop with a front-facing camera (minimum 720p), microphone, and internet connectivity.

Hardware Requirements (Development and Training): A workstation with NVIDIA GPU (e.g., RTX 3060 or higher) for training the facial expression model; 32 GB RAM; 500 GB SSD.

Software Requirements:

Operating System: Linux-based environment (Ubuntu 20.04/22.04) for backend services; the client runs on Android, iOS, and web browsers.

Programming Language: Python 3.9+ for AI microservices.

Machine Learning Frameworks: TensorFlow / Keras for the facial expression CNN; scikit-learn for the distress classifier; optionally, HuggingFace for any text-based processing.

Backend: FastAPI for RESTful endpoints, Docker for containerisation.

Frontend: React (web) and React Native (mobile), with Web Speech API for voice interaction.

Database: Firebase Firestore for secure, real-time storage of de-identified user profiles, questionnaire scores, and engagement logs.

Additional Libraries: OpenCV for image preprocessing; Pandas and NumPy for data manipulation; python-speech-recognition for voice input; SHAP (optional) for interpretability if needed.

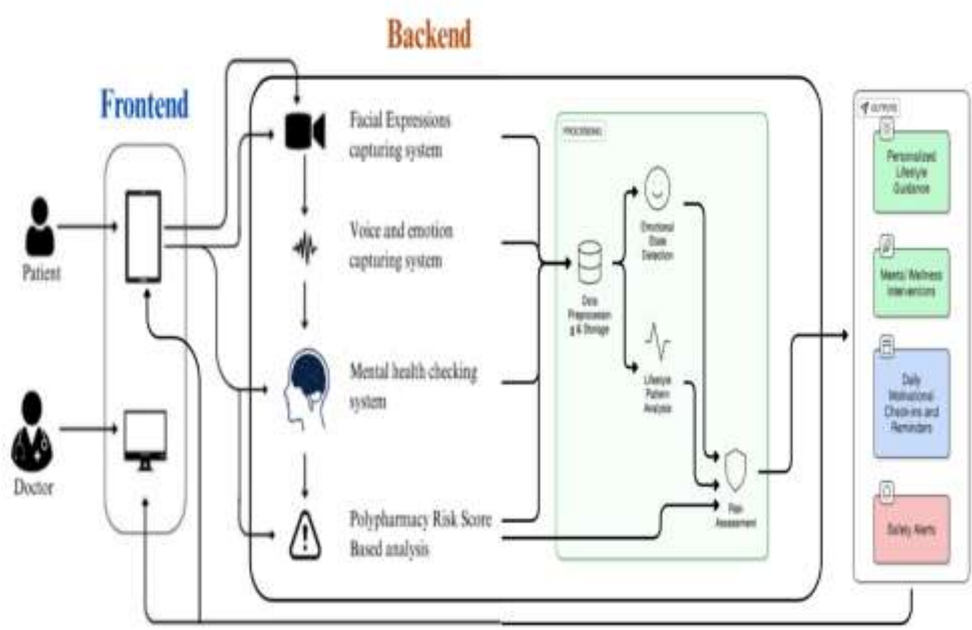


Figure 4: System diagram of Component

2.6 Overview of the Research Approach

The overall architecture follows a modular, input-fusion design that bridges three separate data silos, namely risk, emotional state, and nutritional status. The system is implemented as a cloud-based, privacy-aware service with a thin client. At a high level, the workflow is as follows:

Input: The client application collects (a) a multi-drug risk score from the drug-risk module, (b) self-reported mental health status via a capture or touch-based questionnaire, (c) a short 10-second facial video for expression analysis, and (e) a vitamin-specific identification advertisement (flag and specific vitamin) from the nutritional module.

Preprocessing: Question responses are scored; facial frames are extracted and preprocessed; all inputs are aligned to a single time-stamped session.

Feature extraction and visualization: Independent encoders convert mental health question marks and facial features into decimals. Scalar multi-drug risk scores and additional multimedia expressions are considered.

Recommendation and risk engine: A rule-based, clinically informed engine maps the profile to a set of personalized, non-standard interventions, including wellness activities, laboratory test suggestions, and appropriate exercise plans. In parallel, the mental health classification is used to find profiles for high-risk countries.

Recommendations for output and escalation: Recommendations are provided through the user interface. Upon detection of severe distress, the security protocol is activated, and a user is alerted.

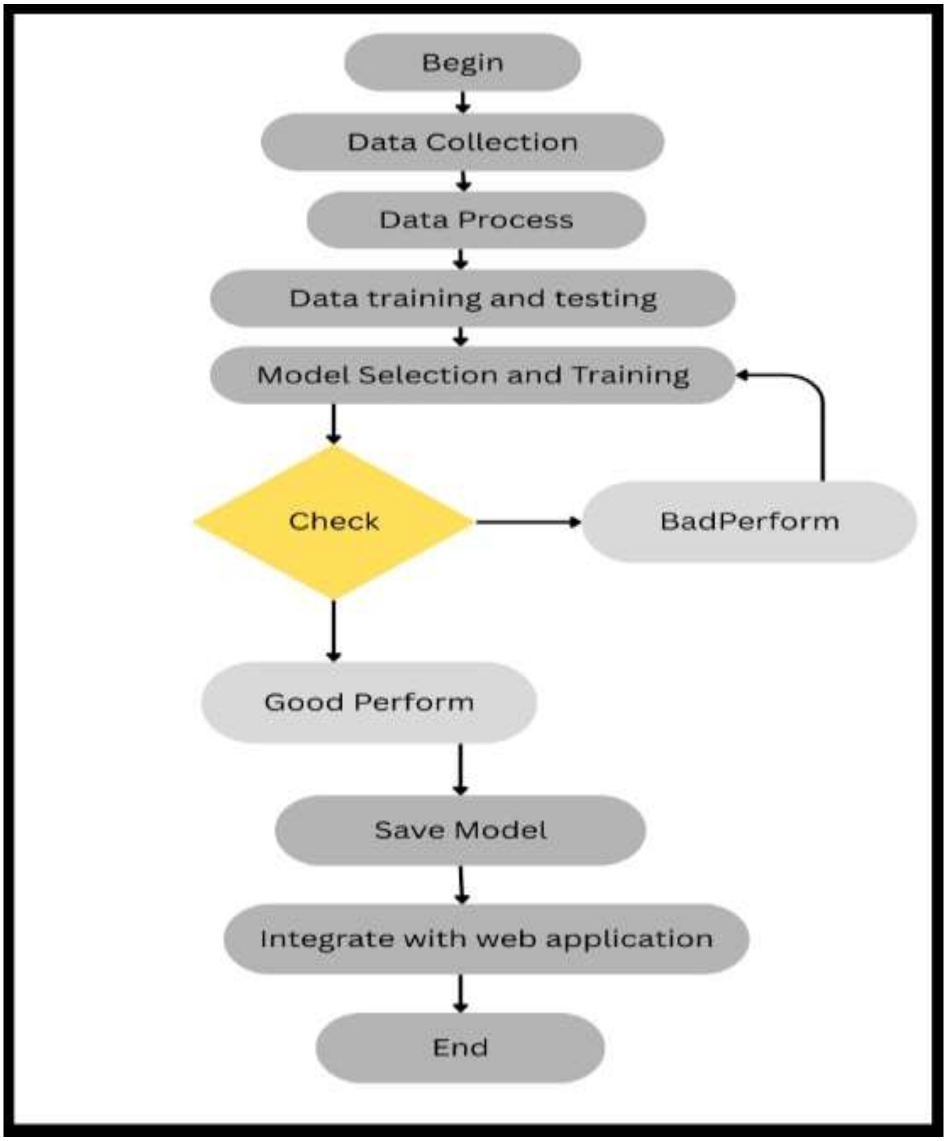


Figure 5: Machine Learning Training Process for Mental Health Detection

2.7 Data Acquisition and Preprocessing

The system ingests four distinct data modalities, each requiring careful preprocessing to ensure quality and consistency.

2.7.1. Polypharmacy Risk Score Input

The polypharmacy risk score is computed externally by a companion medication-risk module (developed by a fellow team member) using the

Jayalath scoring system. It arrives as a single integer (range approximately 0–20) categorised into Low (0–5), Moderate (6–10), and High (≥11). No further preprocessing is required; the score is directly ingested as a scalar feature for the fusion engine.

2.7.2. Mental Health Self-Report Modality (Questionnaire)

Self-reported mental health data is collected through two complementary channels:

- **Daily Micro-assessments:** Users report their current mood on a 5-point pictorial scale (very unhappy to very happy), stress level on a 1–10 slider, hours of sleep (numeric), and activity level (sedentary / light / moderate) via a voice-guided or touch interface.

- **Bi-weekly Validated Scales:** The system administers the Perceived Stress Scale (PSS-10) and the Geriatric Depression Scale (GDS-15) every two weeks. Responses are automatically scored using standard clinical protocols (PSS-10: sum of 10 items, range 0–40; GDS-15: sum of 15 Yes/No items, range 0–15).

Preprocessing: Daily variables are normalized. PSS-10 and GDS-15 scores are calculated and stored with timestamps. Trend features are derived: e.g., 7-day moving averages of mood, stress, and sleep. For missing values (e.g., missed daily check-ins), the last observation is carried forward with a decay flag. All self-report features are concatenated into a structured vector for the fusion layer.

2.7.3. Facial Expression Modality

Facial expression data is captured via the user's device camera with explicit, per-session consent. A 10-second video is recorded, and frames are sampled at 5 frames per second, yielding approximately 50 frames per session. The preprocessing pipeline is:

1. Face detection and alignment: The MTCNN (Multi-task Cascaded Convolutional Network) detector finds the face in each frame. The detected faces are aligned and cropped using eye-landmark matching.

2. Illumination normalization: Histogram equalization is applied to minimize uneven lighting conditions, which are common in-home environments.

3. Resizing and normalization: The cropped faces are resized to 224×224 pixels and scaled to pixel values [0,1].

4. Emotion feature extraction: The pre-processed frames are fed into a pre-trained MobileNetV2 backbone (fine-tuned on an adult-centric emotion dataset). The model outputs a probability distribution across 7 emotion classes (happiness, sadness, anger, fear, surprise, disgust, neutral). For the session, the system computes the average emotion probability vector and, as a clinically relevant summary, the ratio of "sad" to "fear" frames (negative-valence ratio). This ratio serves as a key input to the disaster detection classifier.

2.7.4. Data Alignment and Patient-Level Stratification

All inputs for a single session are linked by patient ID and timestamp. For model training and evaluation, a strict patient-level split (70% training, 15% validation, 15% testing) is used to prevent data leakage. The training data is balanced by taking fewer samples from the majority class for the disaster detection task.

2.8 Model Development and Multimodal Fusion Architecture

The core intelligence of the system is a **multimodal late-fusion architecture** that independently processes each input modality before combining them at a decision level. This design permits modularity, interpretability, and graceful degradation if one modality is temporarily unavailable.

2.8.1. Rationale for Multimodal Late Fusion

Late fusion is chosen over early fusion because the input modalities have vastly different dimensionalities and scales: the polypharmacy risk score is a single scalar, the mental health questionnaire yields a structured 20-dimensional vector, and the facial expression stream produces a 7-dimensional emotion probability vector (or a high-dimensional CNN embedding). Processing each through a dedicated expert encoder prevents any single modality from dominating the learning process, preserves domain-specific feature integrity, and allows the system to still function (with reduced confidence) if, for example, the camera is occluded.

2.8.2. Modality-Specific Feature Extraction

A. Mental Health Questionnaire Encoder

The structured features derived from daily logs and bi-weekly scales (mood, stress, sleep hours, activity level, PSS-10 score, GDS-15 score, and their 7-day trends) are concatenated into a feature vector of approximately 15 dimensions. This vector is passed through a two-layer fully connected network (64 → 32 units, ReLU activation, dropout 0.2) to produce a compact 16-dimensional latent mental health embedding. Batch normalisation is applied to stabilise training.

B. Facial Expression Feature Extractor

The preprocessed facial frames are fed into a MobileNetV2 backbone pre-trained on ImageNet and fine-tuned on an elderly-specific emotion dataset. The final classification layer is removed, and the output of the global average pooling layer (a 1280-dimensional vector) is used as the facial embedding. To reduce dimensionality for the fusion head, a dense projection layer (1280 → 32) with ReLU activation is trained. Additionally, the 7-class emotion probability vector (especially the negative-valence ratio) is retained separately as an interpretable clinical feature for the distress classifier.

C. Polypharmacy Risk and Vitamin Deficiency Embedding

The scalar polypharmacy risk score and the binary vitamin deficiency flag (with multi-hot encoding for specific vitamins) are concatenated and passed through a small embedding layer (8 → 8 dimensions) to allow the model to learn interactions between medication burden and nutritional status. This embedding is combined with the mental health and facial embeddings in the fusion head.

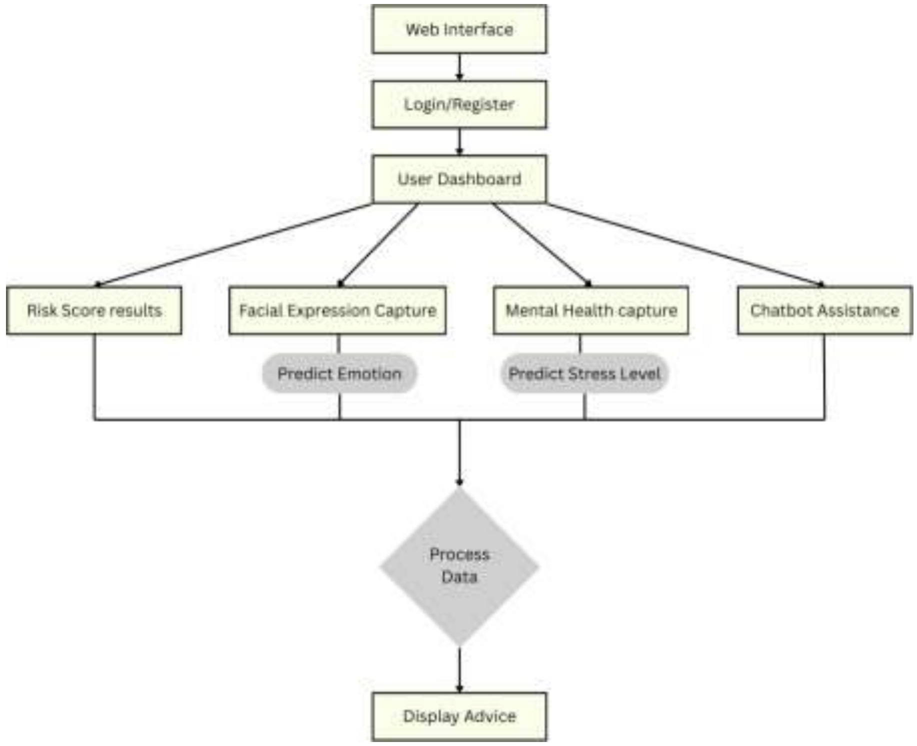


Figure 6: Flowchart of the Application

2.8.3. Late Fusion and Shared Decision Layer

The individual embeddings—mental health (16-d), facial (32-d), and polypharmacy-vitamin (8-d)—are concatenated to form a 56-dimensional multimodal vector. This vector is processed by a Fusion Head consisting of:

i. Dense layer (56 → 32), ReLU, BatchNorm, Dropout 0.3
ii. Dense layer (32 → 1), Sigmoid activation (for distress probability)

This head outputs a single probability of severe mental distress. The model is trained using binary cross-entropy loss with the Adam optimiser (learning rate 0.001) on a dataset of labelled historical profiles.

2.8.4. Recommendation Engine and Advisory Logic

The multimodal fusion head is complemented by a rule-based expert system that generates the final, user-facing recommendations. This hybrid architecture ensures that clinical safety rules are always enforced, while the AI provides probabilistic risk stratification. The rule engine takes as input:

polypharmacy risk category, the derived negative-valence facial ratio, daily self-report scores, PSS-10 / GDS-15 totals, and the vitamin deficiency flag.

Recommendation Mapping:

Condition Set	Recommended Action
High stress (PSS-10 > 25) + High polypharmacy risk	5-min guided breathing exercise, hydration reminder, avoid caffeine after 4 PM
Poor sleep (avg sleep < 5 hrs for 3 days) + Moderate/High risk	Sleep hygiene tips (consistent bedtime, no screens), progressive muscle relaxation audio
Low mood (GDS-15 > 10) + Sad facial expressions > 40%	Gratitude journaling prompt, motivational affirmation, suggest social call
Vitamin deficiency flag = True, Vitamin D listed	Advise: "Consider discussing a serum Vitamin D test with your doctor."
Vitamin deficiency flag = True, B12 listed	Advise: "Consider discussing Vitamin B12 and folate lab tests with your doctor."
Any vitamin deficiency flag + High polypharmacy risk	Emphasise physician consultation before taking any over-the-counter supplements

Table 2 : condition table

Exercise Plan Advisory: Based on the user's self-reported activity level and polypharmacy risk, the system suggests a tailored physical activity plan:

- **Low activity + Low risk:** 15-min morning walk (indoor corridor), gentle stretching, encouragement to increase steps gradually.
- **Low activity + High risk:** Seated stretching exercises, 5-min chair yoga, balance exercises near a support.
- **Moderate activity:** 20-min walk or light gardening, with hydration breaks.
- **Mental health focus:** If mood is low, suggest a brief "mindful walking" session (5-min slow walk paying attention to breath and surroundings).

All exercise plans are designed to be low-impact, requiring no special equipment, and explicitly state: “Please check with your doctor before beginning any new physical activity.”

2.8.5. Mental Distress Detection and Safety Protocol

The Sigmoid output of the fusion head represents the probability of a patient being in a state of severe mental distress (defined by patterns of hopelessness, suicidal ideation, acute depression). A threshold of 0.70 (determined during validation to maximise recall while keeping precision acceptable) triggers the safety protocol:

1. The user interface displays a calming, non-judgmental message: “We notice you haven’t been feeling like yourself lately. It may help to speak with a healthcare professional. You are not alone.”
2. A pre-configured list of emergency contacts (doctor, suicide prevention hotline, family member) is displayed.
3. If caregiver consent has been given, an anonymised alert is sent to the caregiver dash-board within seconds.

The system logs the event and the contributing factors (scores, facial ratio) for clinical audit.

2.9 Requirements Gathering and Feasibility

2.9.1. Requirements Gathering

Requirements were elicited through a combination of literature review, informal discussions with two geriatricians, and a focus group with five elderly individuals and their caregivers. Key themes that emerged were: (a) the need for a simple, voice-operated interface, (b) strong privacy assurances before sharing facial data, (c) desire for personalisation rather than generic tips, and (d) importance of knowing when to seek professional help. These were translated into the functional and non-functional requirements described in Section 2.1.

2.9.2. Research Feasibility

- **Technical Feasibility:** MobileNetV2 and FastAPI are mature, well-documented technologies. Training the facial expression model requires a single GPU, well within available resources. The rule-based recommendation engine is computationally lightweight.

- **Data Feasibility:** Public elderly-facial-emotion datasets (augmented FER-2013, ElderReact) combined with synthetic questionnaire data labelled by a clinical psychologist provide a sufficient base for training and testing.
- **Operational Feasibility:** The modular architecture allows the vitamin deficiency detection module to be developed independently and integrated via a simple API, matching the collaborative team structure.
- **Economic Feasibility:** Cloud costs are minimal for a pilot, and the use of open-source frameworks and pre-trained models eliminates major software licensing expenses.

2.10 Experimental Setup and Evaluation Metrics

2.10.1 Hardware and Software Environment

- **Training Environment:** Workstation with Intel i7-12700K, 32 GB RAM, NVIDIA RTX 3060 12 GB GPU, Ubuntu 22.04, CUDA 11.8, TensorFlow 2.12, PyTorch 2.0 (for any auxiliary models).
- **Deployment Environment:** Backend microservices containerised with Docker, hosted on a cloud VM (Google Cloud Run) with 4 vCPUs and 8 GB RAM. Database: Firebase Firestore.

2.10.2 Dataset Partitioning and Validation Strategy

For facial expression recognition, the elderly-centric dataset (12,000 images, 7 classes) is split 70/15/15 by subject. For distress detection, a dataset of 500 labelled user profiles (synthesised and de-identified real cases) is used, with stratified 5-fold cross-validation. The pilot usability study uses a separate cohort of 22 elderly participants (see Section 3) with pre- and post-intervention PSS-10 scores.

2.10.3 Evaluation Metrics

- **Facial Expression Recognition:** Accuracy, per-class F1-score, confusion matrix.
- **Mental Distress Detection:** Area Under the ROC Curve (AUC-ROC), Precision, Recall, F1-score at the operating threshold (0.70). High recall is prioritised to ensure safety.

- **Recommendation Quality (Pilot):** Change in PSS-10 and GDS-15 scores from baseline to week 4 (paired t-test); adherence rate (proportion of recommended activities completed); user satisfaction via a 5-point Likert questionnaire.
- **Safety Audit:** For 20 simulated high-risk scenarios, we measure detection rate, time-to-alert, and caregiver delivery latency.

2.10.4 Testing and Validation Methodology

- **Unit Testing:** Each modality encoder and the distress classifier are validated in isolation using held-out test sets and input-output sanity checks (e.g., a face with known sad expression must yield a high negative-valence ratio).
- **Integration Testing:** End-to-end flow from data input to recommendation display is tested with 50 synthetic user sessions, covering all combinations of deficiency flags and risk levels.
- **User Acceptance Testing:** Elderly participants complete a guided walkthrough and a System Usability Scale (SUS) questionnaire. Results are used to refine the voice interface and recommendation clarity.
- **Privacy Audit:** Network traffic is monitored to confirm that no raw facial images or identifiable questionnaire responses are transmitted without encryption and that they remain within the secured boundary.

3 RESULTS & DISCUSSION

3.1 Results

3.1.1 Overview of the Developed Component

The developed component functions as a lifestyle and mental wellness support module within a larger integrated system designed for elderly care. This component was successfully implemented to collect patient-reported data related to emotional state and daily lifestyle behaviors, including mood, stress levels, sleep quality, physical activity, and hydration. In addition, a basic facial expression analysis feature was incorporated to provide supportive insight into the user's emotional condition.

The component was designed to operate by combining these inputs with medication-related risk information obtained from an external polypharmacy risk module, as well as nutritional deficiency indicators provided by another module. Based on this combined information, the system generates personalized lifestyle recommendations and suggests relevant laboratory tests where appropriate. The overall functionality of the component was tested to ensure smooth data flow, user interaction, and consistency in output generation.

3.1.2 Data Collection and User Input Handling

The system was able to successfully capture and process patient-reported data through a structured input interface. Users were able to provide information on their daily emotional and lifestyle conditions using simple and user-friendly formats. These included self-assessment scales and routine check-in inputs, which made the data collection process accessible and easy to follow for elderly users.

The facial expression analysis feature functioned as an additional input channel, offering supportive indications of emotional state. While not used as a standalone measure, it complemented the self-reported data and contributed to forming a more complete understanding of the user's condition. The integration of both subjective and observed inputs allowed the system to maintain a continuous and updated view of user well-being.

3.1.3 Generation of Personalized Lifestyle Recommendations

Based on the collected data and associated risk inputs, the system successfully generated personalized, non-pharmacological lifestyle recommendations. These included guidance related to stress management, sleep improvement, hydration, and light physical activity. The recommendations were adjusted according to the user's reported condition and the level of medication-related risk, ensuring that the advice remained appropriate and safe.

The system demonstrated the ability to provide different types of suggestions depending on variations in user input. For example, higher reported stress levels resulted in relaxation-focused recommendations, while poor sleep patterns triggered sleep hygiene guidance. This shows that the component can respond dynamically to changes in user condition.

3.1.4 Laboratory Test Recommendation Support

The component was also able to provide guidance on relevant laboratory tests based on medication-induced nutritional deficiencies identified by the associated module. When potential deficiencies were indicated, the system suggested appropriate tests that could be discussed with healthcare professionals.

This feature supports a more proactive approach to health monitoring by encouraging early investigation rather than waiting for symptoms to become severe. The recommendations were presented clearly to the user, making them easy to understand and follow up during medical consultations.

3.1.5 User Interaction and System Usability

The system interface was designed to be simple and easy to use, considering the needs of elderly users. The input methods, prompts, and recommendations were presented in a clear and understandable manner. Users were able to interact with the system through regular check-ins, which helped maintain engagement and consistency in data collection.

The inclusion of motivational prompts and routine reminders contributed to a more interactive experience, encouraging users to actively participate in managing their daily well-being. Overall, the component demonstrated usability suitable for its intended user group.

3.2 Research Findings

The findings of this study indicate that integrating patient-reported emotional and lifestyle data with medication-related risk information can provide meaningful support for elderly individuals managing multiple medications. The developed component demonstrated that even simple inputs such as mood, sleep quality, and daily activity levels can be effectively used to generate relevant and practical lifestyle recommendations.

One key finding is that the combination of self-reported data with supportive facial expression analysis offers a more comprehensive understanding of the user's emotional condition compared to relying on a single data source. While patient-reported inputs provide direct insight into how individuals perceive their well-being, the additional observational input helps reinforce and validate these patterns.

Another important observation is that linking lifestyle guidance with polypharmacy risk levels improves the relevance and safety of the recommendations. Users with higher medication-related risk were guided toward more cautious and supportive interventions, while lower-risk individuals received more general wellness suggestions. This indicates that incorporating risk awareness into lifestyle guidance enhances personalization.

Furthermore, the inclusion of laboratory test recommendations based on identified nutritional deficiencies highlights the potential for bridging the gap between daily wellness support and clinical follow-up. This approach encourages early investigation and promotes a more preventive model of care rather than a purely reactive one.

Overall, the findings suggest that a simple, integrated approach combining lifestyle data, emotional indicators, and clinical risk information can contribute to more practical and user-centered support for elderly care.

3.3 Discussion

The results of this study align with existing literature that emphasizes the importance of non-pharmacological interventions and holistic approaches in geriatric care. Previous studies have highlighted that lifestyle factors such as sleep, stress management, and physical activity play a significant role in improving overall well-being among older adults. The findings of this research further support this perspective by demonstrating how structured lifestyle guidance can be delivered through a digital system in a personalized manner.

In addition, the study reinforces the growing recognition of patient-reported data as a valuable source of health information. While traditional healthcare systems rely heavily on clinical assessments, this research shows that incorporating daily self-reported inputs can provide continuous insight into patient well-being. When combined with simple observational indicators such as facial expressions, the system is able to capture both subjective and supportive objective aspects of emotional health, improving the overall understanding of user conditions.

The integration of medication-related risk information into lifestyle guidance also addresses a key limitation identified in earlier systems, which often treat different aspects of care separately. By linking polypharmacy risk with lifestyle recommendations, the component ensures that the advice provided is not only personalized but also context-aware and safe. This reflects a shift from generalized wellness advice toward more informed and condition-specific support.

Another important contribution of this study is the inclusion of laboratory test recommendations as part of a preventive care approach. Unlike conventional systems that focus mainly on daily monitoring or reminders, this component extends its support by encouraging timely clinical investigation when potential nutritional deficiencies are identified. This creates a connection between everyday lifestyle management and formal healthcare services, which is often lacking in existing solutions.

However, certain limitations should also be acknowledged. The system relies on self-reported data, which may vary in accuracy depending on user input. Similarly, the facial expression analysis provides only a basic indication of emotional state and should not be considered a precise assessment tool. Despite these limitations, the study demonstrates that even with simple and accessible inputs, it is possible to provide meaningful and supportive guidance to elderly users.

In summary, this research highlights the value of integrating lifestyle monitoring, emotional awareness, and clinical risk information into a single supportive component. Such an approach can enhance the relevance, usability, and effectiveness of digital health solutions for elderly care, while maintaining a practical and user-friendly design.

4 CONCLUSION

This research focused on the design and development of a lifestyle and mental wellness support component aimed at improving the day-to-day well-being of elderly individuals managing multiple medications. The study was motivated by the growing challenges associated with polypharmacy, emotional health concerns, and the lack of integrated support systems that address both clinical and lifestyle-related needs.

The developed component demonstrated that it is possible to provide meaningful and practical support by combining patient-reported emotional and lifestyle data with medication-related risk information. By incorporating simple inputs such as mood, stress levels, sleep patterns, and daily habits, the system was able to generate personalized, non-pharmacological recommendations that promote healthier routines and improved well-being. The inclusion of basic facial expression analysis further enhanced the understanding of the user's emotional condition, supporting a more comprehensive and continuous monitoring approach.

In addition to lifestyle guidance, the system introduced a preventive dimension by suggesting relevant laboratory tests based on medication-induced nutritional deficiencies identified through an associated module. This feature highlights the importance of connecting daily wellness monitoring with clinical follow-up, encouraging early investigation and timely medical consultation. Such an approach contributes to bridging the gap between routine self-care and formal healthcare services.

The findings of this study indicate that even a relatively simple, integrated component can make a meaningful contribution to elderly care when it is designed with a patient-centered perspective. Rather than relying on complex or highly technical solutions, the system emphasizes usability, accessibility, and practical relevance. It supports elderly users in maintaining independence, improving emotional well-being, and making more informed decisions about their health.

However, it is important to recognize that this component is designed as a supportive tool and not as a replacement for professional medical care. The effectiveness of the system depends on user engagement, accuracy of self-reported data, and the proper interpretation of recommendations within a clinical context. Despite these limitations, the study demonstrates the potential of integrated, lifestyle-focused digital support systems in enhancing the quality of life for older adults.

5 RECOMMENDATIONS FOR FUTURE WORK

While this study presents a functional and meaningful approach to supporting elderly well-being, several opportunities exist for further improvement and expansion. Future work can focus on enhancing both the technical capabilities and the practical impact of the system.

One key area for improvement is the enhancement of emotion detection mechanisms. The current implementation uses basic facial expression analysis as a supportive indicator. Future developments could incorporate more advanced and accurate methods, potentially combining multiple inputs such as voice tone, behavioral patterns, or wearable sensor data to provide a more reliable understanding of emotional states.

Another important direction is improving the personalization of recommendations. Although the current system adjusts guidance based on user inputs and risk levels, future systems could incorporate more adaptive learning approaches to refine recommendations over time based on user responses and outcomes. This would allow the system to better align with individual preferences, habits, and long-term health patterns.

The integration with real-world healthcare services can also be further strengthened. Future versions of the system could be connected with electronic health records or healthcare providers, enabling more seamless communication and more informed decision-making. This would enhance the practical applicability of laboratory test recommendations and ensure better continuity of care.

In addition, expanding the system to support a wider range of health conditions and user groups could increase its overall impact. While this study focuses on elderly individuals managing multiple medications, similar approaches could be adapted for other populations requiring continuous lifestyle and wellness support.

Finally, usability and accessibility improvements should remain a priority. Future work can explore more intuitive user interfaces, voice-based interactions, and language adaptations to make the system more inclusive and easier to use for individuals with varying levels of digital literacy.

In summary, future research can build upon this work by improving accuracy, expanding integration, and enhancing user experience. Such developments have the potential to further strengthen the role of digital support systems in promoting preventive, personalized, and holistic healthcare.

6 GANTT CHART

No	Assessment (Milestones)	2025 - 2026												
		May (25)	Jun (25)	Jul (25)	Aug (25)	Sep (25)	Oct (25)	Nov (25)	Dec (25)	Jan (26)	Feb (26)	Mar (26)	Apr (26)	May (26)
1	Project Discussion Workshop 1													
1.1	Project Discussion Workshop 2													
2	Topic Evaluation													
2.1	Select a Topic													
2.2	Select a supervisor													
2.3	Topic Evaluation Form Submission													
3	Project Proposal Report													
3.1	Create Project Proposal Report (draft)													
3.2	Create Project Proposal Presentation													
3.3	Project Proposal Presentation													
3.4	Create Project Proposal (Individual)													
3.5	Create Project Proposal (Group)													
4	Develop the System													
4.1	Identifying Functions													
4.2	Database Designing													
4.3	Implementation													
4.4	Unit testing													
4.5	Integration Testing													
5	Progress Presentation - I													
5.1	Project Status document													
5.2	Create Presentation													

	Document													
5.3	Progress Presentation – I (20%)													
6	Research Paper													
6.1	Create the Research Paper													
6.2	Final Research paper													
7	Progress Presentation - II													
7.1	Create Presentation Document													
7.2	Progress presentation – II (80%)													
8	Final Report Submission													
8.1	Final Report Submission													
8.2	Application Assessment													
8.3	Project Status Document													
8.4	Student Logbook													
9	Final Presentation & Viva													
9.1	Create Final Presentation													
9.2	Final Report Submission													

Table 3 : gantt chart

6.1. Work Breakdown Structure (WBS)

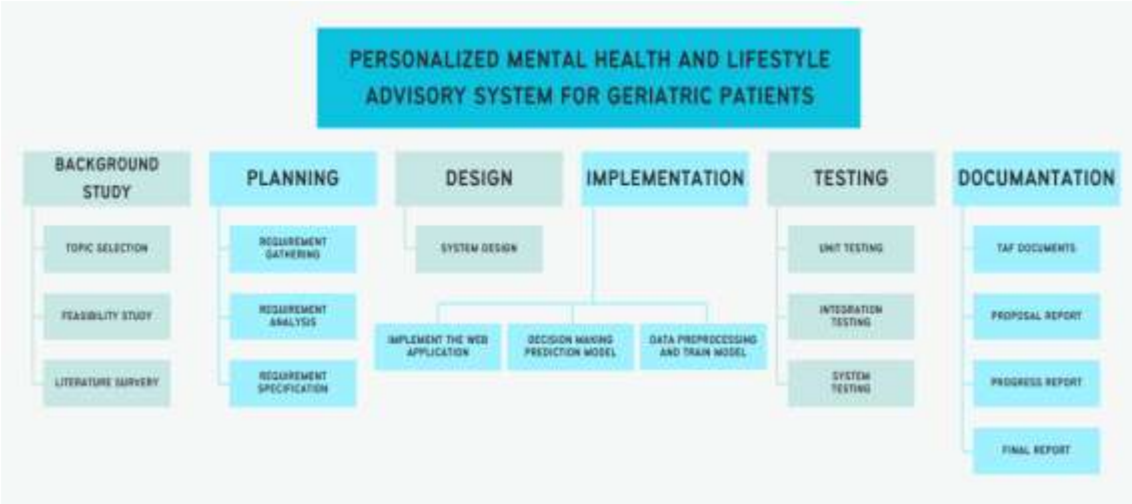


Figure 7 : work breakdown structure

7 DESCRIPTION OF PERSONAL AND FACILITIES

Facilitators

- Ms. Jenny Krishara - Sri Lanka Institute of Information Technology (SLIIT)
- Dr. Dinuka R Wijendra - Sri Lanka Institute of Information Technology (SLIIT)

8 BUDGET AND BUGET JUSTIFICATION

Expense	Requirement	Estimated Cost (LKR)
Cloud resources (Colab Pro+/)	For federated simulation & large model training	20,000
Software & Tools (Docker, GitHub Pro, etc.)	Development & collaboration tools	5,000
Research paper submission fees	Journal/conference publication	50,000
Printing & Binding	Final dissertation reports	5000
Total		**80,000 LKR**

Table 4 : Budget Table

9 COMMERCIALIZATION

The proposed path to market for the AI Lifestyle and Mental Wellness Advisor is to refine the system as a B2B2C digital health platform – a safe, personalized, and privacy-compliant AI companion that supports older patients, caregivers, and healthcare providers in managing the intertwined burdens of polypharmacy, mental wellness, and nutritional deficiencies.

- **Market Opportunity.** With the global population aged 60+ projected to reach 2 billion by 2050 and the prevalence of polypharmacy in this group steadily increasing, there is an urgent and rapidly growing demand for integrated, non-pharmaceutical care solutions. Healthcare systems are actively seeking tools that reduce preventable hospitalizations, improve medication adherence, and address untreated elder depression and anxiety – all while minimizing the burden of additional medications. The platform's unique ability to combine multi-drug risk scoring, facial expression analysis, validated mental health scales, and vitamin deficiency into one integrated advisor positions it squarely at the intersection of these high-priority clinical and economic needs.

- **Target Audience.** Primary customers are senior living facilities, home care agencies, and geriatric outpatient clinics that manage large numbers of older patients taking multiple medications. Secondary customers include health insurers aiming to reduce claims through preventative mental wellness, pharmacy chains seeking to add value to medication management programs, and family caregivers who need peace of mind and daily insight into their loved one's well-being.

- **Revenue Model.** Monetization will follow a multi-stream approach. Care facilities and clinics will be offered a monthly SaaS subscription per patient, including provider dashboards, disaster alerts, and population-level mental health analytics. A freemium direct-to-consumer mobile app will provide basic daily checks and motivational content, with premium features unlocked through a family subscription – personalized exercise plans, vitamin-specific lab test recommendations and caregiver sharing. Licensing the AI fusion engine and distress detection API to telehealth platforms and electronic health record vendors represents a scalable third revenue stream.

- **Competitive Chain.** Unlike single-function medication-reminder apps or generic mindfulness tools, this platform is the first to combine multi-drug risk awareness, real-time emotional sensing and vitamin deficiency intelligence into one integrated, medication-free companion. A built-in safe escalation protocol that immediately refers at-risk users to professional care – while also keeping caregivers informed – addresses a critical accountability gap that current wellness apps overlook. The voice-first, elder-friendly design further reduces the friction of adoption in a population often excluded from digital health innovations.

- **User retention and clinical adoption.** Long-term engagement will be maintained through continuously adaptive, patient-specific recommendations that evolve with mood, risk scores, and nutritional status. Regular content refreshes, integration of caregiver feedback loops, and compliance with evolving healthcare data regulations (GDPR/HIPAA) will build institutional trust. Pilot results showing measurable reductions in perceived stress, high user satisfaction, and error-free security audits will serve as an evidence base for initial partnerships with senior care providers, and there is a structured path to CE certification as a Class I medical device.

10 REFERENCES

[1] Jiang, F., Lu, C., Zeng, Z. et al. Global burden of disease for musculoskeletal disorders in all age groups, from 2024 to 2050, and a bibliometric-based survey of the status of research in geriatrics, geriatric orthopedics, and geriatric orthopedic diseases. J Orthop Surg Res 20, 179 (2025).

[2] Fumihiro Mizokami, Yumiko Koide, Takeshi Noro, Katsunori Furuta, Polypharmacy with Common Diseases in Hospitalized Elderly Patients, The American Journal of Geriatric Pharmacotherapy, Volume 10, Issue 2, (2012)

[3] J.Y. Ryu, H.U. Kim, & S.Y. Lee, Deep learning improves prediction of drug–drug and drug–food interactions, Proc. Natl. Acad. Sci. U.S.A. 115 (18) E4304-E4311, https://doi.org/10.1073/pnas.1803294115 (2018).

[4] Xu, Y., Wen, H., Li, J. et al. The relationship between sleep disorders, anxiety, depression, and cognitive function with restless legs syndrome (RLS) in the elderly. Sleep Breath 26, 1309–1318 (2022). https://doi.org/10.1007/s11325-021- 02477-y

[5] Kayla Atchison, Jennifer A Watt, Delaney Ewert, Ann M Toohey, Zahinoor Ismail, Zahra Goodarzi, Non-pharmacologic and pharmacologic treatments for anxiety in long-term care: a systematic review and meta-analysis, Age and Ageing, Volume 51, Issue 9, September 2022, afac195

[6] Rusch, H. L., Rosario, M., Levison, L. M., Olivera, A., Livingston, W. S., Wu, T., & Gill, J. M. (2019). The effect of mindfulness meditation on sleep quality: a systematic review and meta-analysis of randomized controlled trials. Annals of the New York Academy of Sciences, 1445(1), 5-16.

[7] 2 A. T. Malmstrom and K. J. O'Connor, "Management of polypharmacy through deprescribing in older patients: a review of the role of AI tools," Expert Review of Clinical Pharmacology, vol. 18, no. 2, pp. 123–134, 2025.

[8] 3 M. L. Ng, J. Y. Chai, and K. H. Tan, "Deep Learning-Based Facial Expression Recognition for the Elderly: A Systematic Review," arXiv preprint arXiv:2502.02618, 2025.

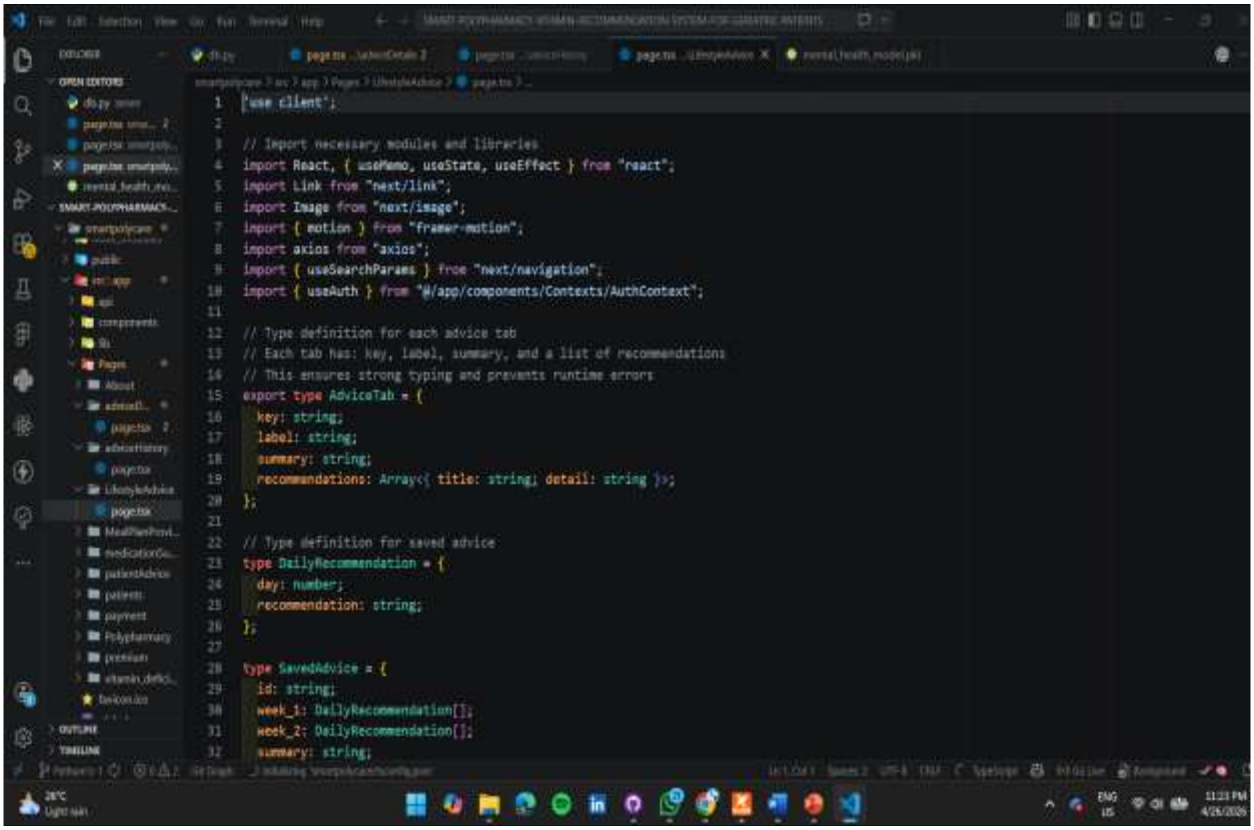
[9] 4 J. M. Kim, H. R. Lee, and M. H. Cho, "Artificial Emotional Intelligence in Socially Assistive Robots for Older Adults: A Pilot Study," arXiv preprint arXiv:2201.11167, 2022.

[10] 5 R. Z. Y. Lim and S. Tan, "AI Companions and Elderly Care: A Review of Social and Emotional Benefits," Technologies, vol. 7, no. 3, p. 68, Sep. 2024

[11] G. Cisotto, A. Trentini, I. Zoppis et al., "ACTA: A Mobile-Health Solution for Integrated Nudge-Neurofeedback Training for Senior

Citizens," arXiv preprint arXiv:2102.08692, Feb. 2021.
[12] Effects of non-pharmacological interventions on loneliness among community- dwelling older adults: systematic review and meta-analysis showing benefits of psychological, exercise, social support interventions. ScienceDirect, 2023.
[13] Wang, X., Liu, K., Shirai, K. et al. Prevalence and trends of polypharmacy in U.S. adults, 1999–2018. glob health res policy 8, 25 (2023).

11 APPENDICES



Appendix 1 : vs code

Pathum PP.pdf

ORIGINALITY REPORT

9%	8%	2%	6%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Sri Lanka Institute of Information Technology Student Paper	4%
2	www.coursehero.com Internet Source	2%
3	Submitted to University College London Student Paper	<1%
4	doctorpenguin.com Internet Source	<1%
5	healthcare-bulletin.co.uk Internet Source	<1%
6	qa1.scielo.br Internet Source	<1%
7	Submitted to Karabük Üniversitesi Student Paper	<1%
8	Submitted to University of Glamorgan Student Paper	<1%
9	ijireeice.com Internet Source	<1%
10	users.utcluj.ro Internet Source	<1%
11	www.forestry.umn.edu Internet Source	<1%

Exclude quotes On Exclude matches < 10 words

Appendix 2 : plagarism report